ALEXANDER DSOUZA

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PROFESSIONAL SUMMARY

Principal Engineer with 15+ years building and scaling software systems, specializing in Python and DevOps. Proven track record architecting distributed data systems and leading cross-functional teams, with strong DevOps and reliability excellence across cloud infrastructure. AI-first engineer skilled in prompt engineering and LLM integration. Recognized Top External Contributor to Mozilla CI.

TECHNICAL SKILLS

- **Backend:** Python, Django, FastAPI, Django REST Framework (DRF), Pydantic, Node.js, SQL
- **Frontend:** React, Redux, React Router, Vite, Material UI (MUI), TypeScript
- **Infrastructure & DevOps:** Kubernetes (k3s/GKE), Docker, AWS, Ansible, Helm, CI/CD (GitHub Actions, GitLab)
- **Observability:** Prometheus, Grafana, Jaeger, Loki, OpenTelemetry, New Relic, Snyk
- **Core Expertise:** Microservices, SDN, Distributed Systems, API Design (Protobuf/JSON), Message Queues (RabbitMQ), SOC 2 Compliance
- **AI/ML:** Prompt Engineering, LLM Integration, Daily GenAI-Assisted Development (Cursor, Aider, Jules, Copilot), Machine Learning Fundamentals

PROFESSIONAL EXPERIENCE

Independent Contributor — Open Source & Backend Engineering Practice
Oct 2025 – Present

Self-Directed | Pune, India

- Completed the AWS AI/ML Scholar Challenge 2026 (Udacity); received multiple mentions on planet.mozilla.org throughout 2026 for contributions to Treeherder and PerfCompare.
- Deepened expertise in FastAPI and Django, and built hands-on proficiency in React, TypeScript, and Redux via component and state-management work on PerfCompare.
- Advanced PostgreSQL/Redis via query optimization, indexing, and scaling patterns.

Principal Software Engineer
Apr 2018 – Oct 2025

Jeavio India Pvt Ltd | Pune, India

Promoted twice internally over 7.5 years: Senior QA Engineer → Senior Software Engineer → Principal Engineer.

- Led cross-functional teams delivering SDN and WiFi collectors (async, multithreaded Python, high-throughput JSON processing) for IBM SevOne NMS across multiple releases, for enterprise customers including BT; owned sprint planning and daily scrums.
- Introduced structured on-call rotation and documented RCA practices to prevent burnout and speed up fixes, and drove a Shift-Left QA approach that increased unit test coverage and product quality.
- Led CI/CD strategy across repositories and the customer installer, building multi-stage GitLab pipelines integrated with vSphere, JFrog, and Slack, then migrating to GitHub Actions as part of the 2023 IBM transition; directed containerization/deployment via Docker, Helm, and Ansible.
- Delivered the Versa SD-WAN collector as Senior Software Engineer, then oversaw the collectors' rebuild in Golang and managed the SevOne-to-IBM team transition, mentoring interns on CI/CD ownership.
- Led full-stack design of a document-handling system for a global dairy enterprise (owning the DRF API layer and directing frontend architecture) and orchestrated the AWS architecture upgrade that achieved SOC 2 compliance.

Quality Assurance Engineer II
Feb 2016 – Feb 2018

Amazon India Pvt Ltd | Chennai, India

- Developed a high-performance Protobuf-to-JSON SerDe for global ingestion pipelines and automated validation for high-scale address data; published a tutorial on it (medium.com/@moijes12).

Senior Test Engineer
Mar 2014 – Feb 2016

Honeywell Automation India | Pune, India

- Architected a custom performance measurement test framework in Python; led feasibility analysis for mobile automation, selecting Appium.

Software Test Engineer
Feb 2012 – Feb 2014

Avaya India | Pune, India

- Served as Test Lead for the One Touch Video and Social Media Manager platforms; administered virtualized test environments (VMware ESX) and automated database tasks with Python.

Project Engineer
Jan 2008 – Sep 2011

Wipro Technologies | India

- Handled end-to-end testing and automation for Nokia Base Station Controller GPRS systems.

OPEN SOURCE & KEY ACHIEVEMENTS

- Mozilla CI Top Contributor: merged 11+ PRs to Treeherder and PerfCompare, working with RabbitMQ-based Pulse task/job queueing, React, React Router, and TypeScript.
- Legacy Resolution: resolved 21+ legacy Mozilla issues in Treeherder. Hackathon Winner: designed an on-demand environment pipeline that reduced setup time for issue reproduction. Mentored junior developers and interns into full-time roles.

EDUCATION

Bachelor of Engineering, Information Technology

Mumbai University (FCRIT)